

DEEP LEARNING FOR PERCEPTION

Vision-Based Food Calorie Estimation

Using Object Detection and Multi-Dataset Classification

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1. Task Definition

This project addresses the problem of automated food calorie estimation from images. Traditional calorie tracking is manual, error-prone, and requires nutritional expertise. Our system provides an end-to-end deep learning pipeline capable of identifying food items in real-world plate photographs and estimating their caloric content.

The pipeline consists of two sequential stages:

- Stage 1 — Food Detection:** A YOLOv8m model detects food regions in an image, producing bounding box annotations that localize all food items on a plate regardless of category.
- Stage 2 — Food Classification:** Each detected crop is passed to an EfficientNet-B0 classifier fine-tuned on 748 food classes. The predicted category is mapped to its nutritional profile to produce a per-item and total calorie estimate.

2. Dataset Description

2.1 Object Detection Datasets

Two publicly available datasets were combined to build the detection training corpus:

Dataset	Format	Label Type	Purpose
UEC Food 256	Images	Bounding boxes	Food localization
Open Images v7	Images	Bounding boxes	Diverse food scenes

The detection dataset uses a single-class annotation scheme (class: food), with YOLO-format bounding box labels structured into images/train, images/val, labels/train, and labels/val directories.

2.2 Classification Datasets

Four large-scale food classification datasets were merged to create a comprehensive 748-class taxonomy:

Dataset	Original Classes	Format
Food-101	101	Images (JPEG)
Khana	81	Images (JPEG)
ISIA Food-500	500	Images (JPEG)
FoodX-251	251	Images (JPEG)

After merging, identical or semantically equivalent classes across datasets were collapsed into a single canonical class, yielding 748 unique food categories. Each class contains between 250 and 2,000 images after filtering.

3. Data Pre-processing

3.1 Detection Pre-processing

Images were resized to 640×640 pixels — YOLOv8's standard input resolution. Built-in Mosaic augmentation, random flips, and scale jitter were enabled via the Ultralytics framework defaults.

3.2 Classification Pre-processing

The following augmentation and normalization pipeline was applied:

Step	Operation	Parameters
Crop (Train)	RandomResizedCrop	Output: 224×224
Flip (Train)	RandomHorizontalFlip	p=0.5
Color Jitter (Train)	ColorJitter	brightness=0.3, contrast=0.3, saturation=0.3, hue=0.1
Rotation (Train)	RandomRotation	$\pm 20^\circ$
Grayscale (Train)	RandomGrayscale	p=0.05
Resize (Val)	Resize + CenterCrop	256 $\rightarrow$ 224×224
Normalize (Both)	Normalize	$\mu=[0.485,0.456,0.406]$, $\sigma=[0.229,0.224,0.225]$

3.3 Class Merging and Size Filtering

Class merging was performed manually. Class names across all four datasets were examined for semantic equivalence (e.g., 'french_fries', 'fries', 'French Fries' were unified). Naming convention differences — underscores, capitalization, language — were resolved and a canonical label assigned. A per-class size constraint of 250–2,000 images was imposed: classes below the minimum were discarded; excess images beyond 2,000 were randomly subsampled. A WeightedRandomSampler was additionally applied during training to handle residual class imbalance.

4. Network Architecture

4.1 Detection — YOLOv8m

YOLOv8m is a single-stage, anchor-free detector operating on a multi-scale feature pyramid with three components:

- **Backbone (CSPDarknet):** Extracts hierarchical spatial features using C2f modules, improving gradient flow over the older C3 modules.
- **Neck (PAN-FPN):** Combines multi-scale features via top-down and bottom-up pathways, enabling robust detection of food items at varying sizes.
- **Head (Decoupled):** Separates classification and box regression. Coordinates are predicted anchor-free via Distribution Focal Loss using a task-aligned assignment strategy. The model was initialized from ImageNet weights (yolov8m.pt) and trained on one class: food.

4.2 Classification — EfficientNet-B0

EfficientNet-B0 (via timm, ImageNet pre-trained) uses Mobile Inverted Bottleneck (MBConv) blocks with Squeeze-and-Excitation attention. Compound scaling balances depth, width, and resolution, making B0 the most parameter-efficient variant. The final head was replaced with a linear layer outputting 748 logits; the full network was fine-tuned end-to-end for 20 epochs (~5.3 M parameters).

Component	Detection (YOLOv8m)	Classification (EfficientNet-B0)
Backbone	CSPDarknet + C2f	MBConv + SE attention
Input Size	640×640	224×224
Output	Bounding boxes (1 class)	Class probabilities (748 classes)
Pre-training	ImageNet (yolov8m.pt)	ImageNet (timm)
Parameters	~25.9M	~5.3M

5. Loss Function

5.1 Detection Loss (YOLOv8)

YOLOv8 uses a three-term composite loss:

- **Classification Loss (BCE):** Binary Cross-Entropy with sigmoid activation. Weight: 0.5.
- **Box Regression Loss (CIoU + DFL):** Complete IoU for direct box overlap optimization combined with Distribution Focal Loss for coordinate prediction. Weight: 7.5.
- **DFL Loss:** Additional distribution focal term for anchor-free regression. Weight: 1.5.

$$\text{Total Loss} = 0.5 \times L_{\text{cls}} + 7.5 \times (L_{\text{CIoU}} + L_{\text{DFL}}) + 1.5 \times L_{\text{DFL}}$$

5.2 Classification Loss (EfficientNet-B0)

Cross-Entropy with label smoothing ($\epsilon = 0.1$) was used. This softens hard targets from $\{0, 1\}$ to $\{\epsilon/K, 1 - \epsilon + \epsilon/K\}$ where $K = 748$, preventing overconfident predictions and improving generalization — particularly beneficial over Focal Loss for large class counts.

6. Hyperparameters

Hyperparameter	Detection (YOLOv8m)	Classification (EfficientNet-B0)
Optimizer	SGD (Ultralytics default)	AdamW
Learning Rate	0.01 (initial)	1e-3 (initial)
LR Scheduler	Cosine Annealing (built-in)	CosineAnnealingLR (T_max=20, $\eta_{\min}$ =1e-6)
Batch Size	32 (multi-GPU)	128
Epochs	50	20
Image Size	640×640	224×224 (train), 256 → 224 (val)
Weight Decay	5e-4 (default)	1e-4
Early Stopping	patience=10	Not applied; best model saved
GPU	Multi-GPU (device='0,1')	Kaggle T4

Selection Methodology: YOLOv8m default hyperparameters from the Ultralytics framework were used for detection. For classification, AdamW with $\text{lr}=1\text{e-}3$ follows recommended practice for fine-tuning EfficientNet. Cosine Annealing was chosen over StepLR for smoother decay; `label_smoothing=0.1` was selected for its documented superiority in large-class tasks.

7. SOTA Comparison

7.1 Detection — Our Result

Our YOLOv8m model was trained and evaluated on a custom food detection dataset (UEC Food 256 + Open Images v7), not on COCO. The best checkpoint was obtained at epoch 37. Results are reported on the food-domain validation set:

Metric	Value (Best Epoch 37)
mAP@0.5	62.93%
mAP@0.5:0.95	44.89%
Precision	72.83%
Recall	55.35%

7.1.1 Comparison with Published COCO Benchmarks

The table below shows published results of comparable architectures on the standard COCO benchmark (mAP@0.5:0.95). These numbers are not directly comparable to our result since COCO is a general-purpose 80-class dataset while our task is single-class food detection. The comparison is provided purely for architectural context.

Model	mAP@0.5:0.95 (COCO)	Params	Domain
YOLOv5m	45.2%	21.2M	COCO
YOLOv8m (pub.)	50.2%	25.9M	COCO
YOLOv9m	51.4%	20.0M	COCO
RT-DETR-R50	53.1%	42M	COCO
YOLOv8m (Ours)	44.89% (mAP@0.5:0.95) / 62.93% (mAP@0.5)	25.9M	Food (custom)

7.2 Classification — Our Result

Our EfficientNet-B0 classifier was trained on a merged 748-class food dataset — a significantly harder task than any published single-dataset benchmark. The validation accuracy achieved after 20 epochs is reported below:

Note: The training was halted at 13 epochs, the best accuracy was achieved at epoch 6, so that model is integrated in the final system

Metric	Value
Validation Accuracy	64.51%
Number of Classes	748
Training Epochs	20
Architecture	EfficientNet-B0
Dataset	Food-101 + Khana + ISIA Food-500 + FoodX-251 (merged)

7.2.1 Comparison with Published Classification Benchmarks

All published baselines below are evaluated on Food-101 (101 classes). Our task has 748 classes including regionally specific, visually similar, and underrepresented categories — making direct accuracy comparison inappropriate. The table is provided for architectural reference only.

Method	Top-1 Acc	Classes	Dataset
DeepFood (2016)	77.4%	101	Food-101
ResNet-50 fine-tuned	88.0%	101	Food-101
EfficientNet-B0 (pub.)	~88%	101	Food-101
ViT-B/16 fine-tuned	92.1%	101	Food-101
EfficientNet-B0 (Ours)	64.51%	748	Custom merged

8. Qualitative Discussion

The detection model achieved mAP@0.5 of 62.93% and mAP@0.5:0.95 of 44.89% on the food-domain validation set. The classifier reached 64.51% top-1 accuracy across 748 classes. The lower accuracy relative to Food-101 published results is expected: our merged dataset spans 7.4x more classes, includes visually similar regional dishes, and contains inherent label noise from the merging process.